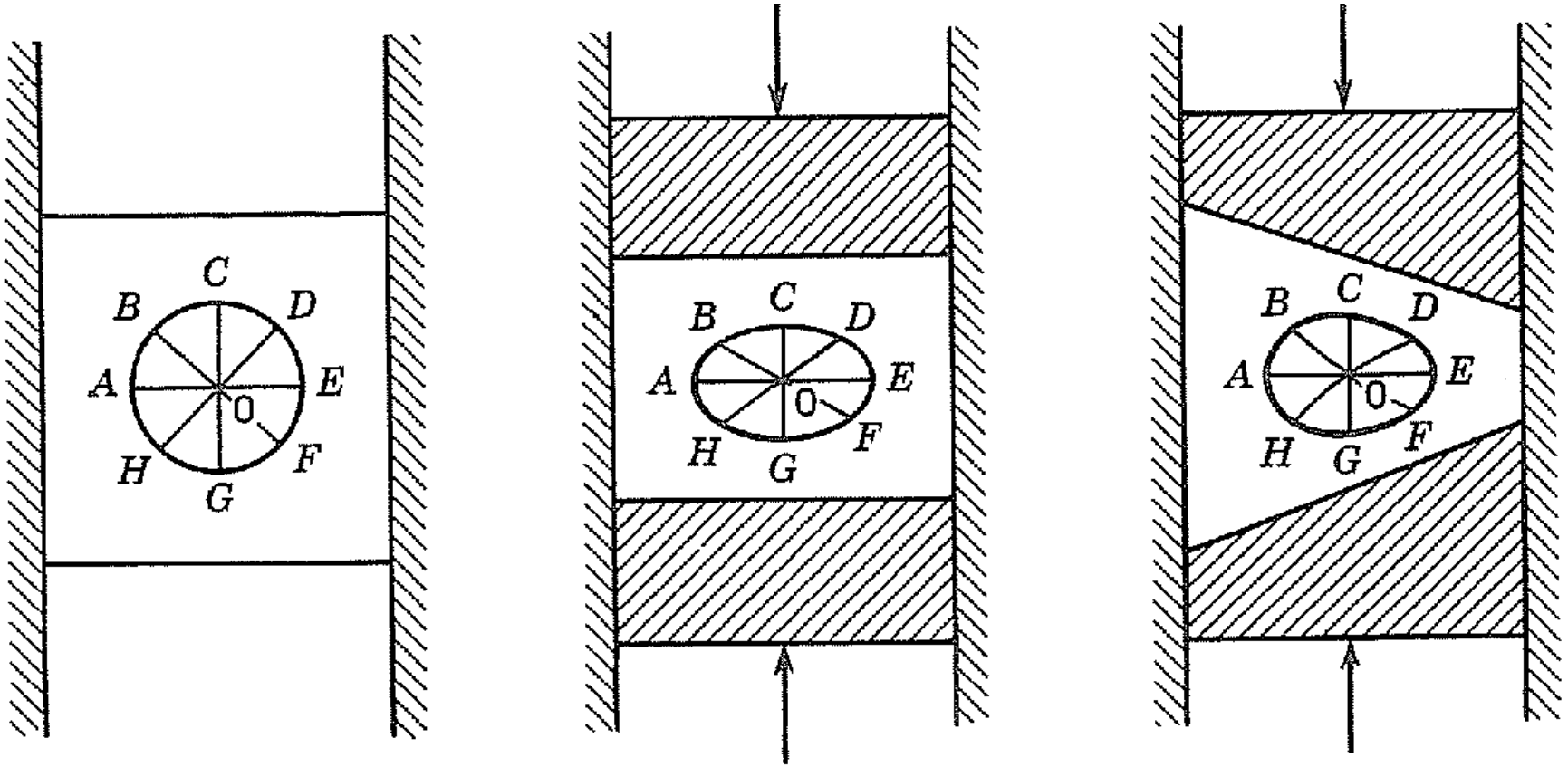
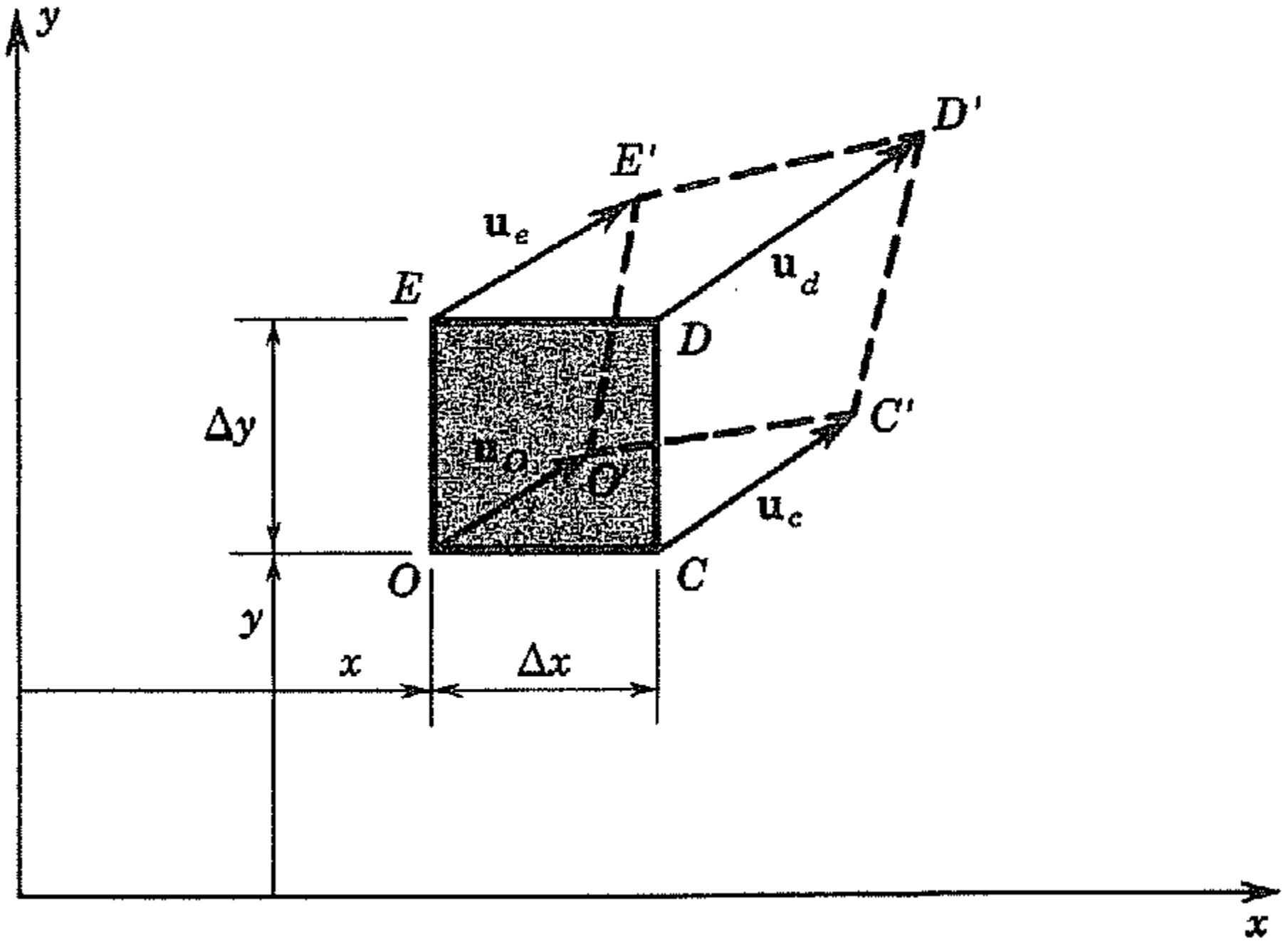
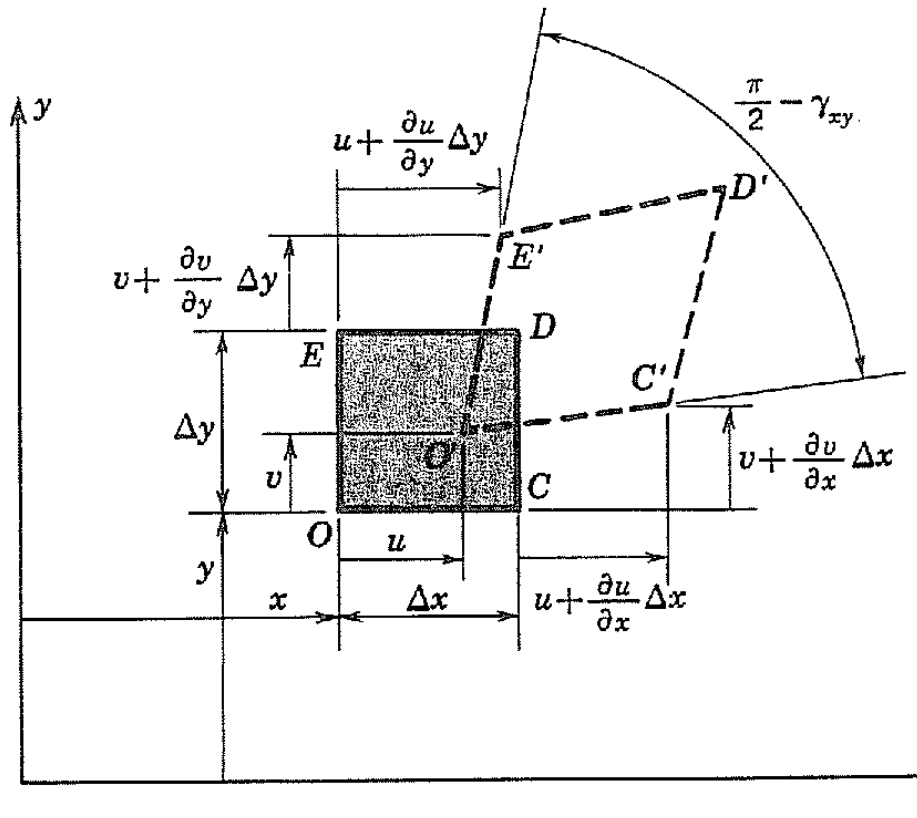
Definition of the strain components




y
D'
E'
$\mathbf{u}_e$
E
$\mathbf{u}_d$
D
Δy
C'
$\mathbf{u}_o$
O
$\mathbf{u}_c$
O
C
y
x
Δx
x

Relation between strain and displacement in plane strain



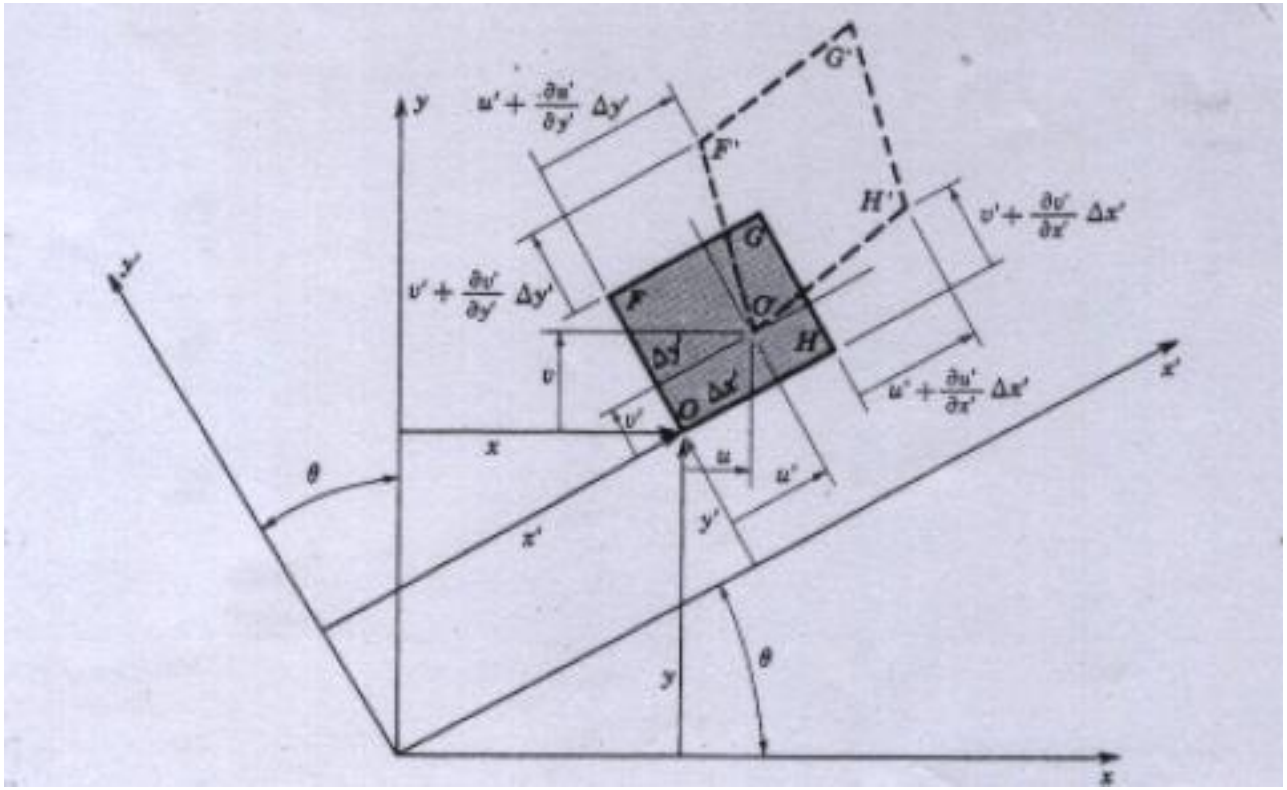
$$\epsilon_x = \lim_{\Delta x \to 0} \frac{O'C' - OC}{OC} = \lim_{\Delta x \to 0} \frac{[\Delta x + (\partial u/\partial x)\Delta x] - \Delta x}{\Delta x} = \frac{\partial u}{\partial x}$$

$$\epsilon_y = \lim_{\Delta y \to 0} \frac{O'E' - OE}{OE} = \lim_{\Delta y \to 0} \frac{[\Delta y + (\partial v/\partial y)\Delta y] - \Delta y}{\Delta y} = \frac{\partial v}{\partial y}$$

$$\gamma_{xy} = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left(\frac{\pi}{2} - \angle C'O'E'\right) = \lim_{\substack{\Delta x \to 0 \\ \Delta y \to 0}} \left\{\frac{\pi}{2} - \left[\frac{\pi}{2} - \frac{(\partial v/\partial x)\Delta x}{\Delta x} - \frac{(\partial u/\partial y)\Delta y}{\Delta y}\right]\right\}$$

$$= \frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}$$

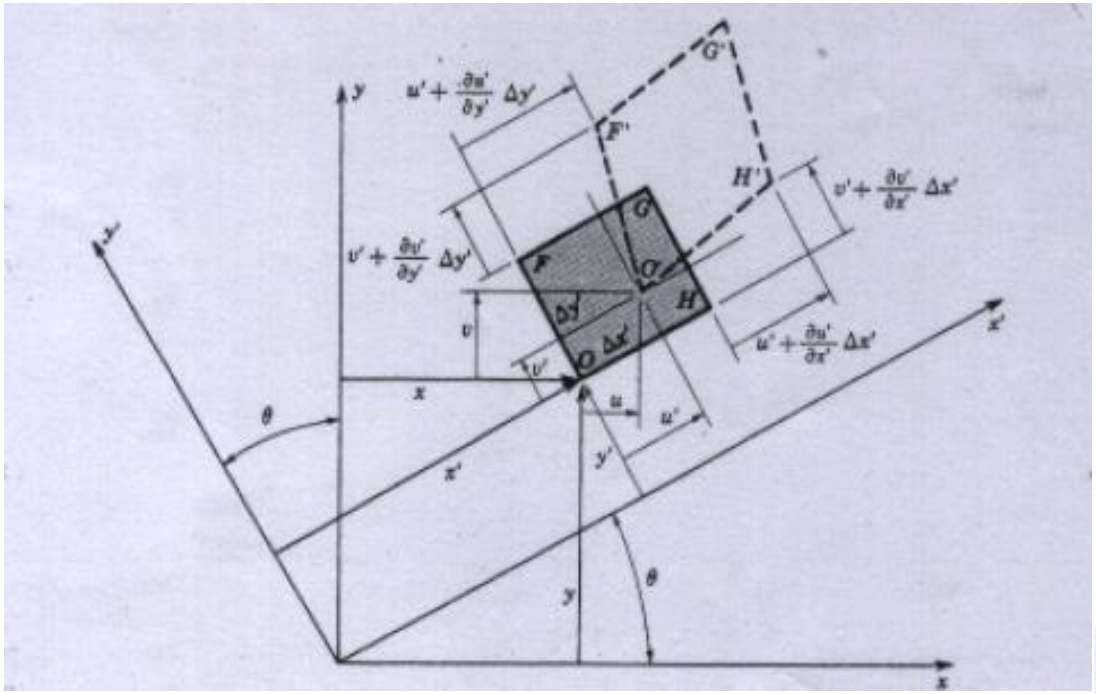
Strain components associated with arbitrary set of axes



$$\epsilon_{x'} = \frac{\partial u'}{\partial x'}$$

$$\epsilon_{y'} = \frac{\partial v'}{\partial y'}$$

$$\gamma_{x'y'} = \frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'}$$



$$\epsilon_{x'} = \frac{\partial u'}{\partial x'} = \frac{\partial u'}{\partial x}\frac{\partial x}{\partial x'} + \frac{\partial u'}{\partial y}\frac{\partial y}{\partial x'}$$

$$\epsilon_{y'} = \frac{\partial v'}{\partial y'} = \frac{\partial v'}{\partial x}\frac{\partial x}{\partial y'} + \frac{\partial v'}{\partial y}\frac{\partial y}{\partial y'}$$

$$\gamma_{x'y'} = \frac{\partial v'}{\partial x'} + \frac{\partial u'}{\partial y'} = \left(\frac{\partial v'}{\partial x}\frac{\partial x}{\partial x'} + \frac{\partial v'}{\partial y}\frac{\partial y}{\partial x'}\right) + \left(\frac{\partial u'}{\partial x}\frac{\partial x}{\partial y'} + \frac{\partial u'}{\partial y}\frac{\partial y}{\partial y'}\right)$$

$$x = x' \cos\theta - y' \sin\theta$$
$$y = x' \sin\theta + y' \cos\theta$$
$$u' = u \cos\theta + v \sin\theta$$
$$v' = -u \sin\theta + v \cos\theta$$

$$\epsilon_{x'} = \left(\frac{\partial u}{\partial x}\cos\theta + \frac{\partial v}{\partial x}\sin\theta\right)\cos\theta + \left(\frac{\partial u}{\partial y}\cos\theta + \frac{\partial v}{\partial y}\sin\theta\right)\sin\theta$$
$$= \frac{\partial u}{\partial x}\cos^2\theta + \frac{\partial v}{\partial y}\sin^2\theta + \left(\frac{\partial v}{\partial x} + \frac{\partial u}{\partial y}\right)\sin\theta\cos\theta$$

$$\epsilon_{x'} = \epsilon_x \cos^2\theta + \epsilon_y \sin^2\theta + \gamma_{xy} \sin\theta\cos\theta$$

$$\epsilon_{x'} = \frac{\epsilon_x + \epsilon_y}{2} + \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta + \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\epsilon_{y'} = \frac{\epsilon_x + \epsilon_y}{2} - \frac{\epsilon_x - \epsilon_y}{2}\cos 2\theta - \frac{\gamma_{xy}}{2}\sin 2\theta$$

$$\frac{\gamma_{x'y'}}{2} = -\frac{\epsilon_x - \epsilon_y}{2}\sin 2\theta + \frac{\gamma_{xy}}{2}\cos 2\theta$$

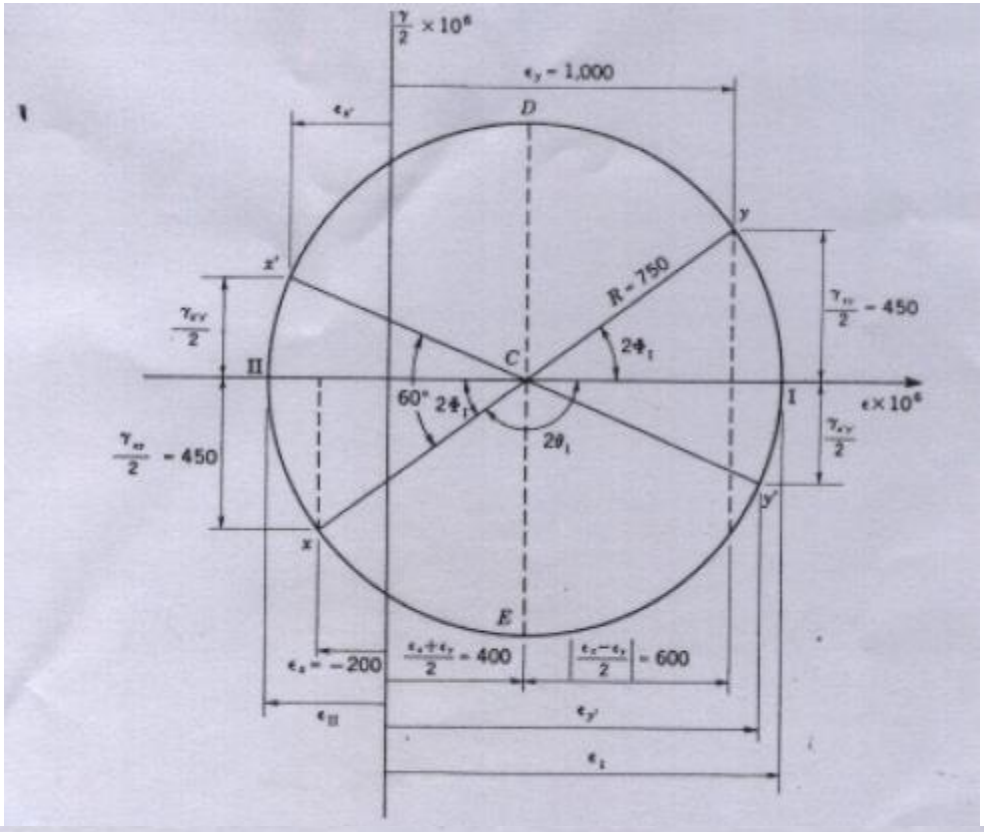
A sheet of metal is deformed uniformly in its own plane so that the strain components related to a set of axes xy are[1]

$$\epsilon_x = -200 \times 10^{-6}$$

$$\epsilon_y = 1000 \times 10^{-6}$$

$$\gamma_{xy} = 900 \times 10^{-6}$$

We wish to find the strain components associated with a set of axes $x'y'$ inclined at an angle of 30° clockwise to the xy set. Also, we wish to find the principal strains and the direction of the axes on which they exist.



$$R = \sqrt{600^2 + 450^2} = 750$$

$$\epsilon_{x'} = (400 \times 10^{-6}) - (750 \times 10^{-6}) \cos (60° - 36.8°) = -290 \times 10^{-6}$$

$$\epsilon_{y'} = (400 \times 10^{-6}) + (750 \times 10^{-6}) \cos (60° - 36.8°) = 1{,}090 \times 10^{-6}$$

Because point x' lies above the ϵ axis (and point y' below), the shear strain $\gamma_{x'y'}$ is negative.

$$\frac{\gamma_{x'y'}}{2} = -(750 \times 10^{-6}) \sin (60° - 36.8°) = -295 \times 10^{-6}$$

$$\gamma_{x'y'} = -590 \times 10^{-6}$$

The principal strains are

$$\epsilon_I = (400 \times 10^{-6}) + (750 \times 10^{-6}) = 1{,}150 \times 10^{-6}$$

$$\epsilon_{II} = (400 \times 10^{-6}) - (750 \times 10^{-6}) = -350 \times 10^{-6}$$

4.20. In a state of plane strain in the xy plane the strain components associated with the xy axes are

$$\epsilon_x = 800 \times 10^{-6}$$
$$\epsilon_y = 100 \times 10^{-6}$$
$$\gamma_{xy} = -800 \times 10^{-6}$$

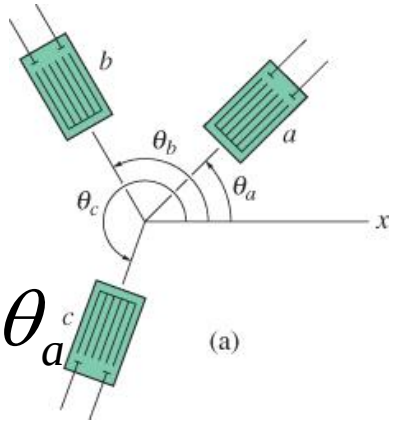
Find the magnitude of the principal strains and the orientation of the principal strain directions.

STRAIN ROSETTES

- We measure the normal strain in a tension-test specimen using an electrical-resistance strain gauge.
- For general loading on a body, the normal strains at a pt are measured using a cluster of 3 electrical-resistance strain gauges.
- Such strain gauges, arranged in a specific pattern are called strain rosettes.

STRAIN ROSETTES

- Apply strain transformation to each gauge:



$$\varepsilon_a = \varepsilon_x \cos^2\theta_a + \varepsilon_y \sin^2\theta_a + \gamma_{xy} \sin\theta_a \cos\theta_a$$

$$\varepsilon_b = \varepsilon_x \cos^2\theta_b + \varepsilon_y \sin^2\theta_b + \gamma_{xy} \sin\theta_b \cos\theta_b$$

$$\varepsilon_c = \varepsilon_x \cos^2\theta_c + \varepsilon_y \sin^2\theta_c + \gamma_{xy} \sin\theta_c \cos\theta_c$$

- We determine the values of ε_x, ε_y γ_{xy} by solving the three equations simultaneously.

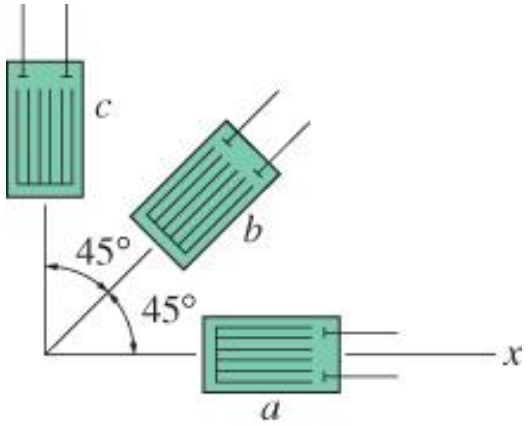
STRAIN ROSETTES

- For rosettes arranged in the 45° pattern, Eqn becomes

$$\varepsilon_x = \varepsilon_a$$

$$\varepsilon_y = \varepsilon_c$$

$$\gamma_{xy} = 2\varepsilon_b - (\varepsilon_a + \varepsilon_c)$$



45° strain rosette

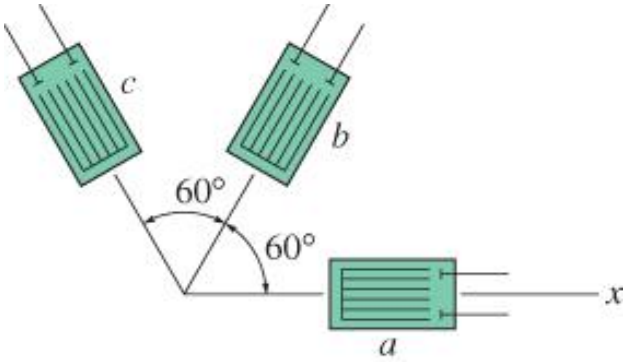
(b)

- For rosettes arranged in the 60° pattern, Eqn becomes

$$\varepsilon_x = \varepsilon_a$$

$$\varepsilon_y = \frac{1}{3}(2\varepsilon_b + 2\varepsilon_c - \varepsilon_a)$$

$$\gamma_{xy} = \frac{2}{\sqrt{3}}(\varepsilon_b - \varepsilon_c)$$

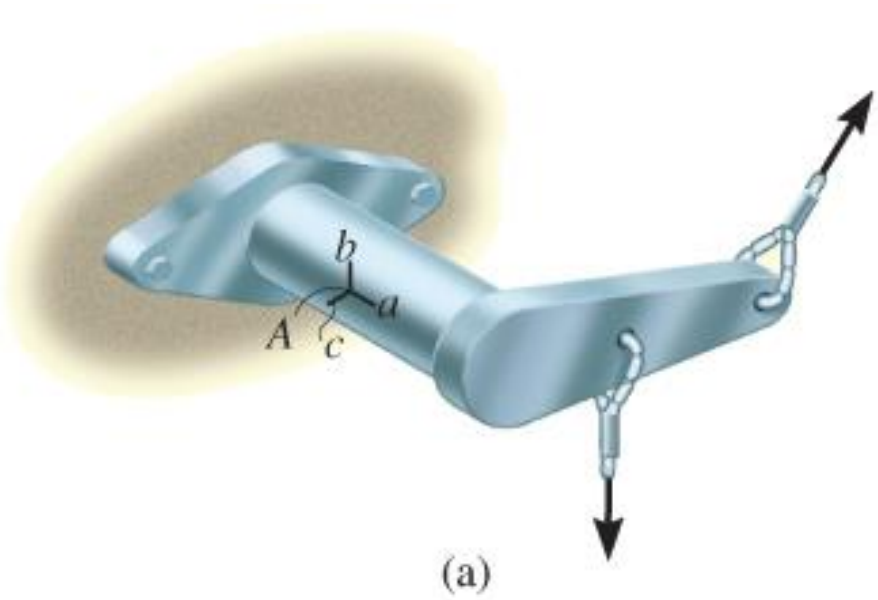


60° strain rosette

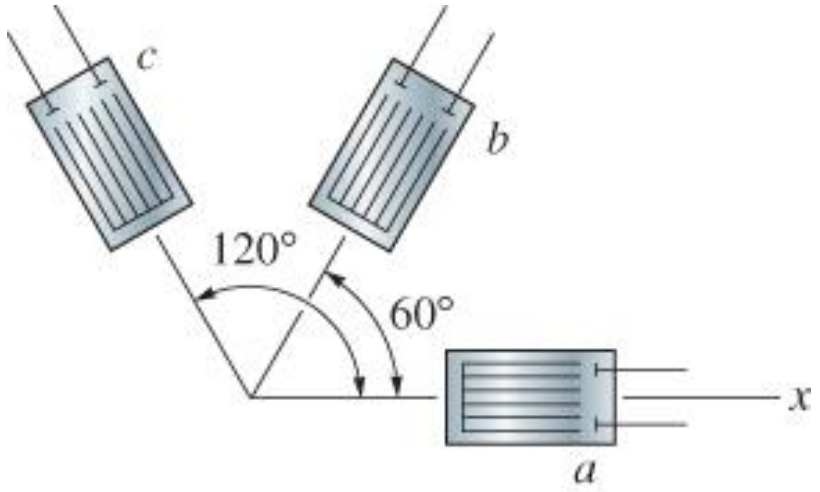
(c)

EXAMPLE

State of strain at pt A on bracket is measured using the strain rosette shown. Due to the loadings, the readings from the gauges give $\varepsilon_a = 60(10^{-6})$,
$\varepsilon_b = 135(10^{-6})$, and $\varepsilon_c = 264(10^{-6})$. Determine the
in-plane principal strains at the pt and the directions in which they act.



(a)



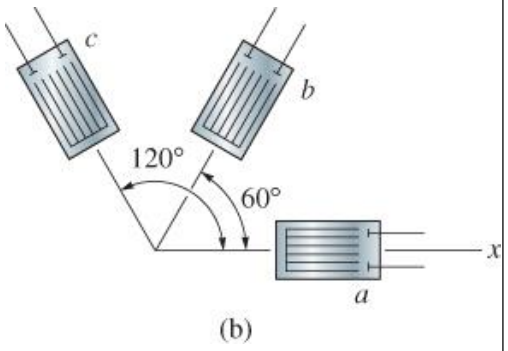
(b)

EXAMPLE 10.8 (SOLN)

Establish x axis as shown, measure the angles counterclockwise from the $+x$ axis to center-lines of each gauge, we have

$\theta_a = 0°$, $\theta_b = 60°$, and $\theta_c = 120°$

Substitute into Eqn 10-16,



$$60\left(10^{-6}\right) = \varepsilon_x \cos^2 0° + \varepsilon_y \sin^2 0° + \gamma_{xy} \sin 0° \cos 0° = \varepsilon_x \qquad (1)$$

$$135\left(10^{-6}\right) = \varepsilon_x \cos^2 60° + \varepsilon_y \sin^2 60° + \gamma_{xy} \sin 60° \cos 60°$$

$$= 0.25\varepsilon_x + 0.75\varepsilon_y + 0.433\gamma_{xy} \qquad (2)$$

$$264\left(10^{-6}\right) = \varepsilon_x \cos^2 120° + \varepsilon_y \sin^2 120° + \gamma_{xy} \sin 120° \cos 120°$$

$$= 0.25\varepsilon_x + 0.75\varepsilon_y - 0.433\gamma_{xy} \qquad (3)$$

EXAMPLE 10.8 (SOLN)

Solving Eqns (1), (2) and (3) simultaneously, we get

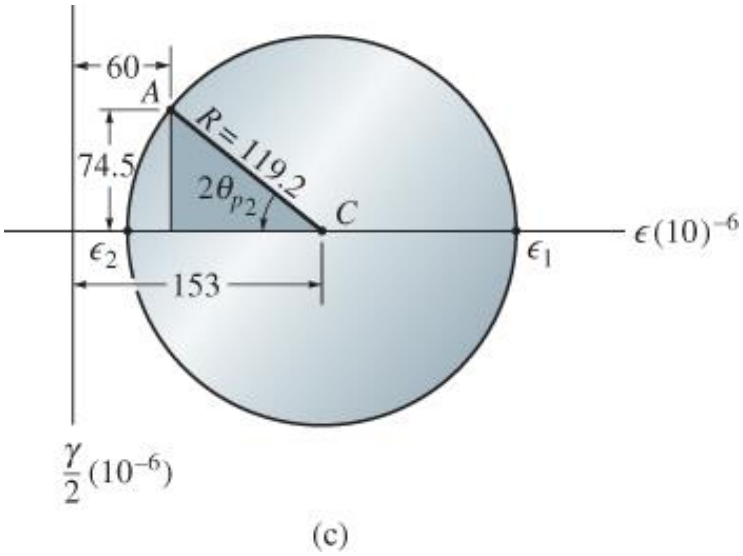
$$\varepsilon_x = 60\left(10^{-6}\right) \qquad \varepsilon_y = 246\left(10^{-6}\right) \qquad \gamma_{xy} = -149\left(10^{-6}\right)$$

The in-plane principal strains can also be obtained directly from Eqn 10-17. Reference pt on Mohr's circle is A [60(10^{-6}), –74.5(10^{-6})] and center of circle, C is on the ε axis at $\varepsilon_{avg} = 153(10^{-6})$.

From shaded triangle, radius is

$$R = \left[\sqrt{(153-60)^2 + (74.5)^2}\right]\left(10^{-6}\right)$$

$$R = 119.2\left(10^{-6}\right)$$



(c)

EXAMPLE 10.8 (SOLN)

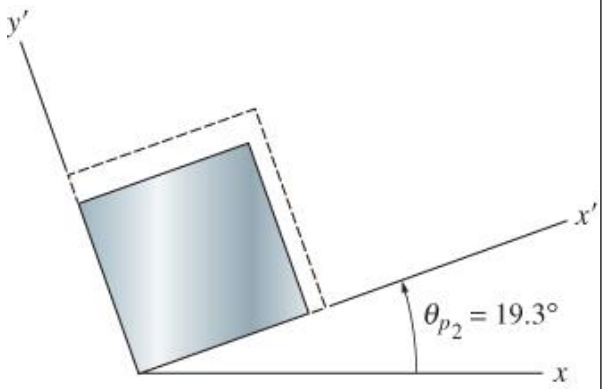
The in-plane principal strains are thus

$$\varepsilon_1 = 153\left(10^{-6}\right) + 119.2\left(10^{-6}\right) = 272\left(10^{-6}\right)$$

$$\varepsilon_2 = 246\left(10^{-6}\right) - 119.2\left(10^{-6}\right) = 33.8\left(10^{-6}\right)$$

$$2\theta_{p2} = \tan^{-1}\frac{74.5}{(153-60)} = 38.7°$$

$$\theta_{p2} = 19.3°$$



(d)